
MODEL SELECTION AND VALIDATION REPORT (general_kde_feve model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (general_kde_feve model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: student(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	0.06	0.09
sbt_mean	-0.06	0.04
sbt_sd	-0.09	0.03
sst_sd	0.02	0.02
sic_mean	-0.06	0.04
log_chla_mean	0.00	0.02
depth	-0.04	0.03

Dispersion parameter: 0.64

Spatiotemporal AR1 correlation (rho): 0.83

Matérn range: 253.00

Spatiotemporal marginal AR1 SD: 0.48

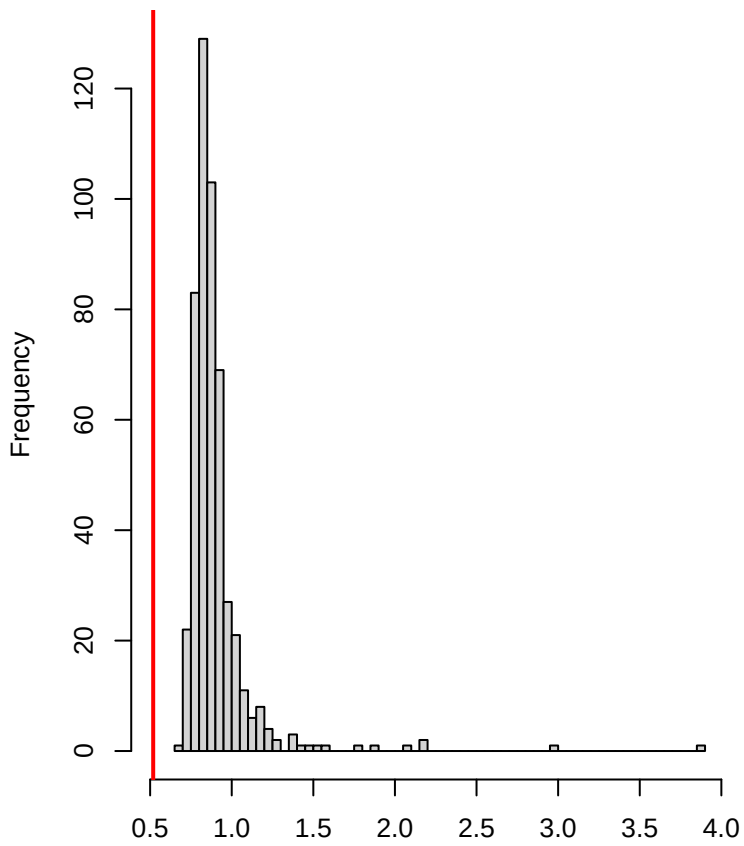
REML criterion at convergence: 4611.376

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

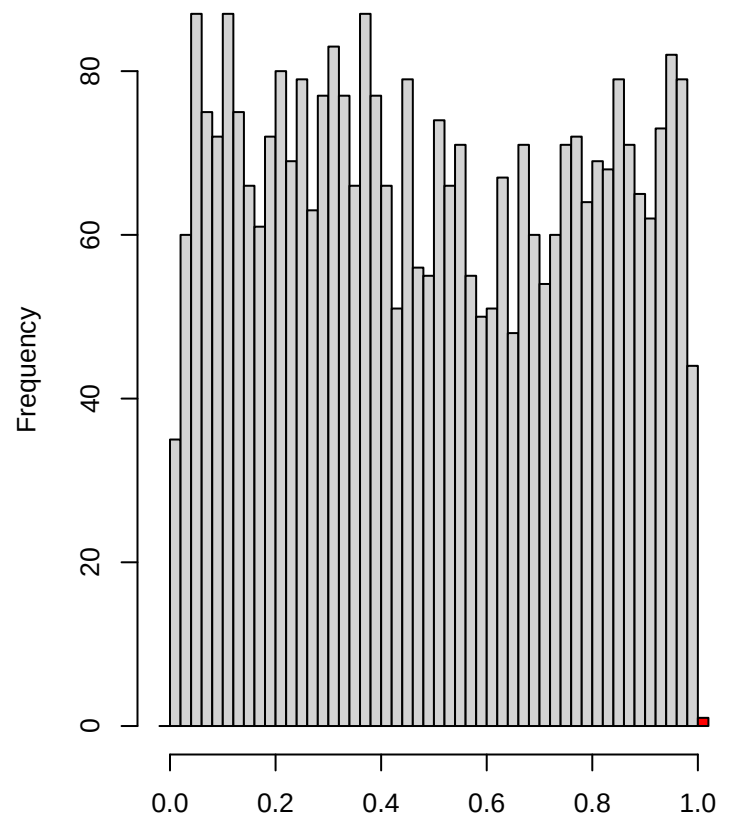
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1_with_anisotropy	13	9244.281	0.4702431
spatiotemporal_random_fields_ar1	11	9244.752	93.9727326
spatiotemporal_random_fields_iid_with_anisotropy	12	9338.724	8.9446275
spatiotemporal_random_fields_iid	10	9347.669	54.2486631
spatial_random_fields	10	9401.918	3.5455994
spatial_random_fields_with_anisotropy	12	9405.463	214.7620646
error_only	8	9620.225	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



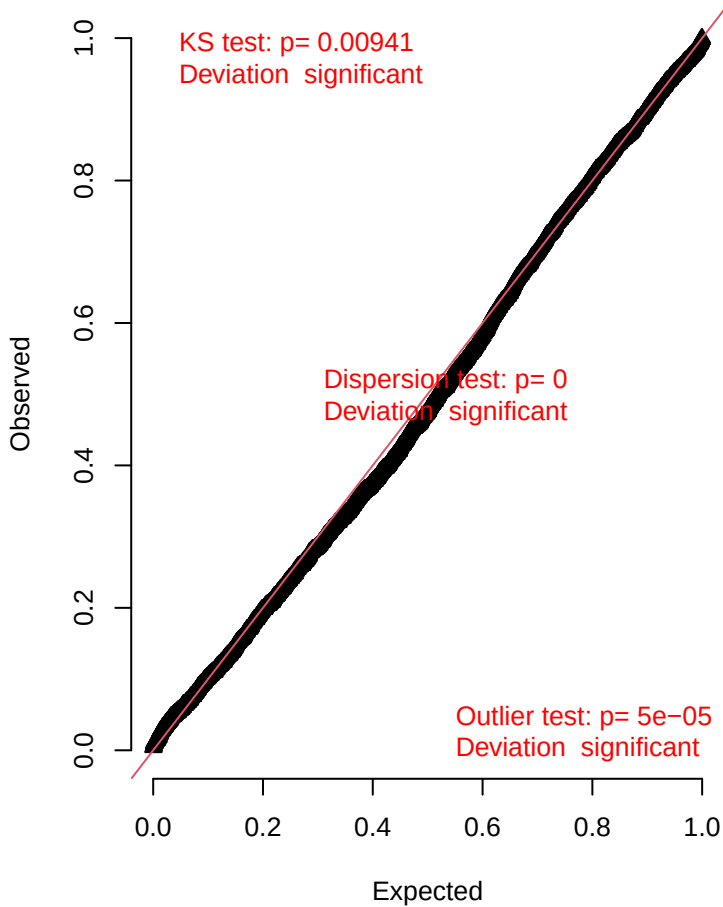
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

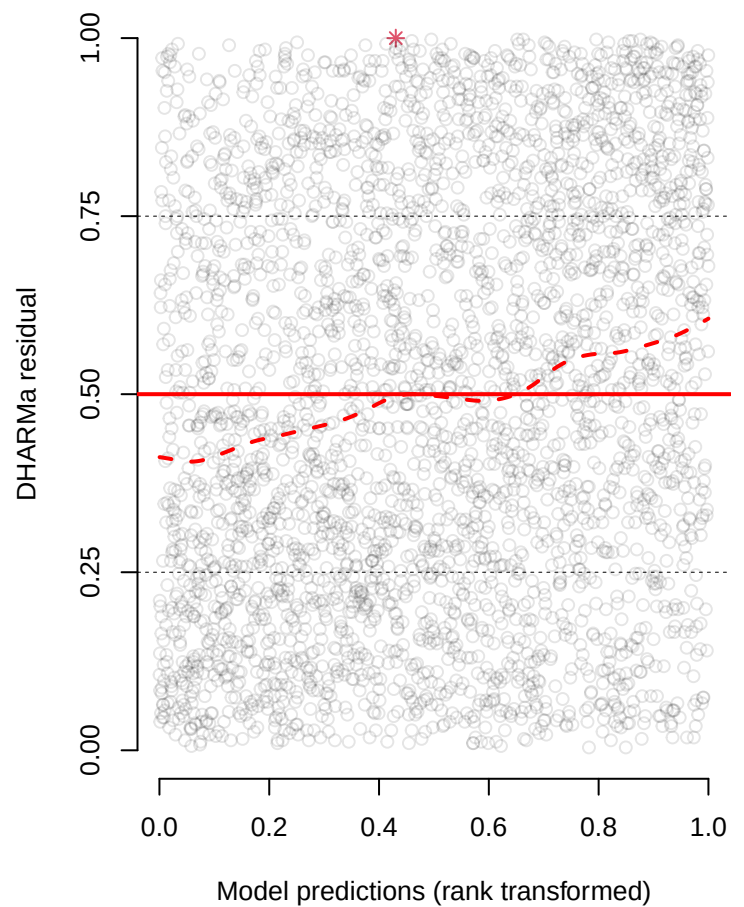


Residuals (outliers are marked red)

QQ plot residuals

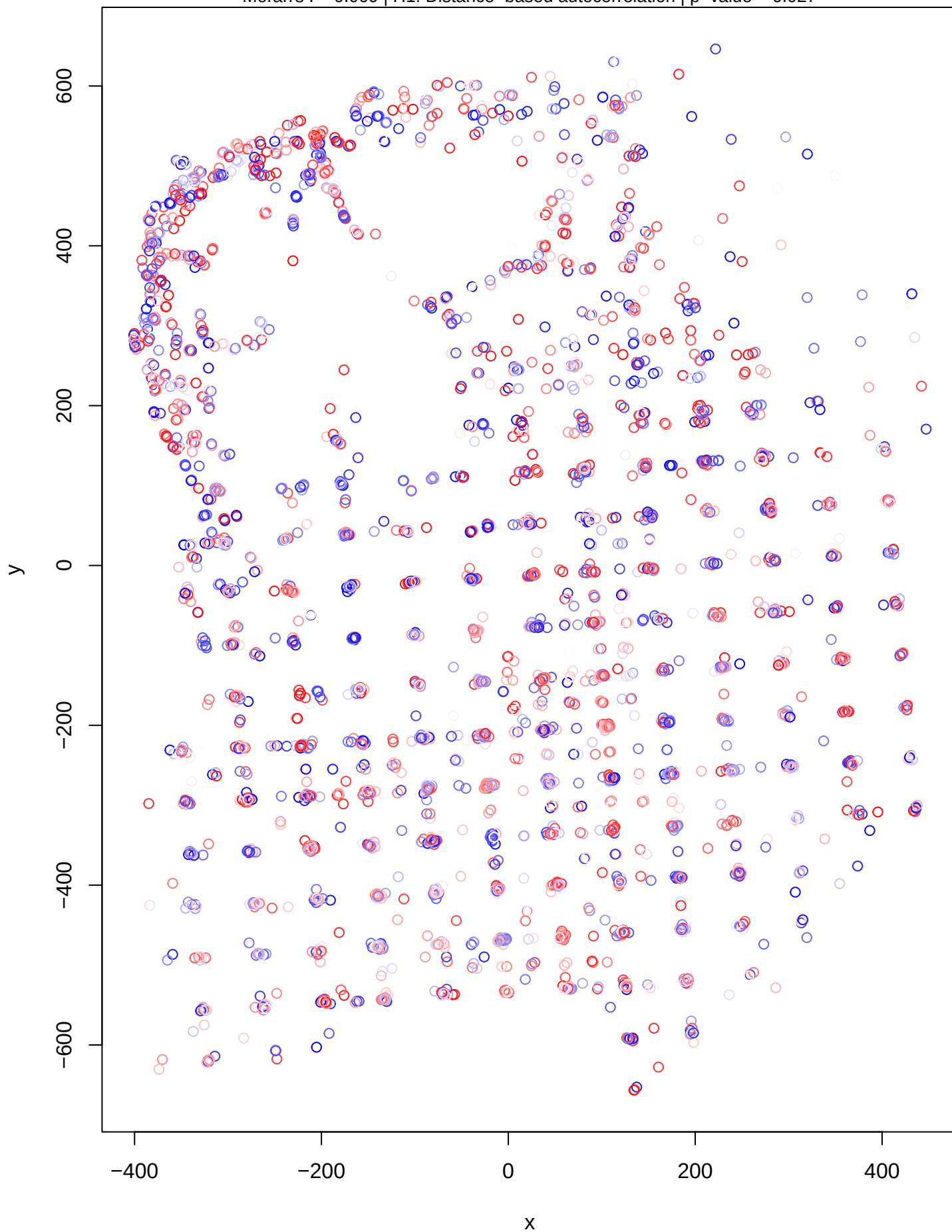


DHARMa residual vs. predicted



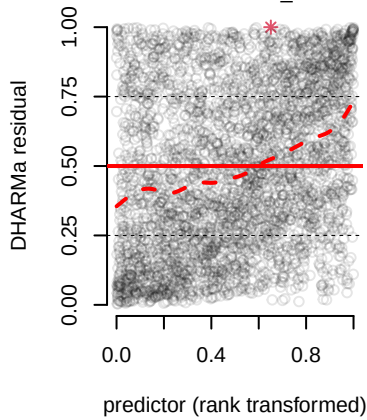
DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.009 | H1: Distance-based autocorrelation | p-value = 0.027



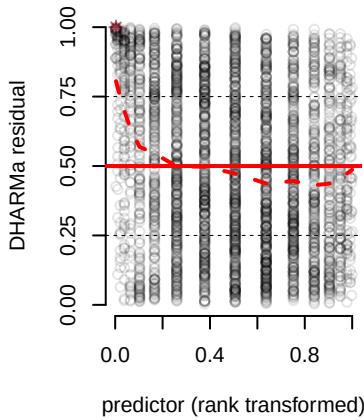
DHARMa residual Residual vs. pred

Predictor: kde_fric



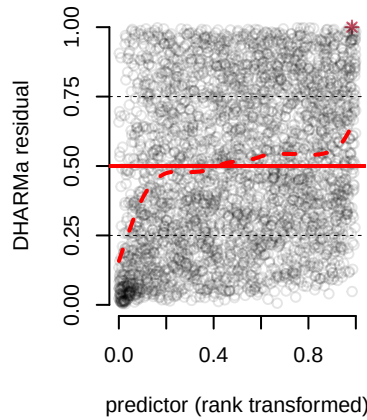
DHARMa residual Residual vs. pred

Predictor: tric



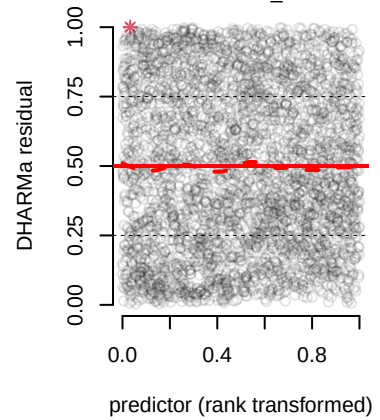
DHARMa residual Residual vs. pred

Predictor: teve



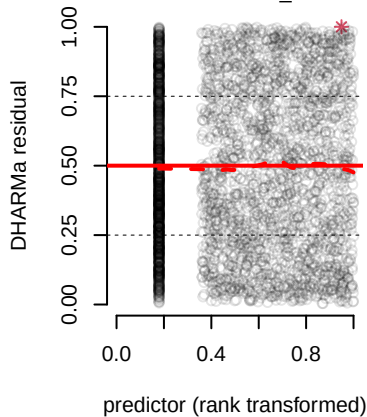
DHARMa residual Residual vs. pred

Predictor: sst_mean



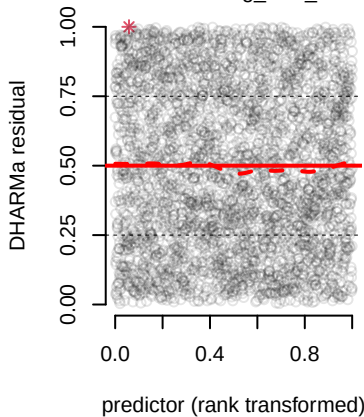
DHARMa residual Residual vs. pred

Predictor: sic_sd



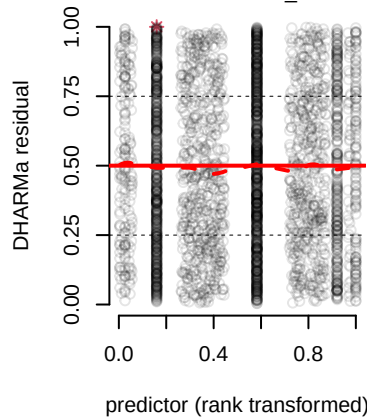
DHARMa residual Residual vs. pred

Predictor: log_chla_sd



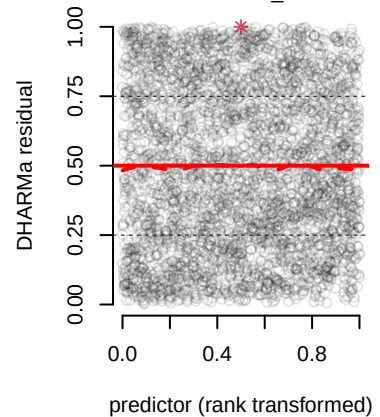
DHARMa residual Residual vs. pred

Predictor: haul_dur



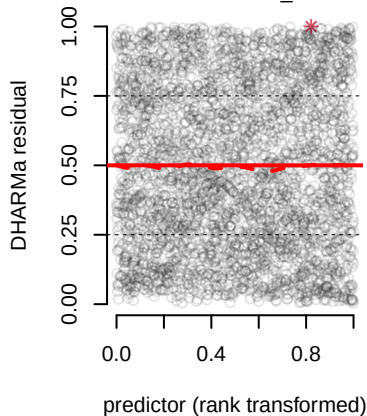
DHARMa residual Residual vs. pred

Predictor: sbt_mean



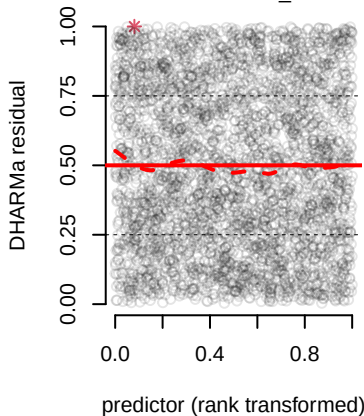
DHARMa residual Residual vs. pred

Predictor: sbt_sd



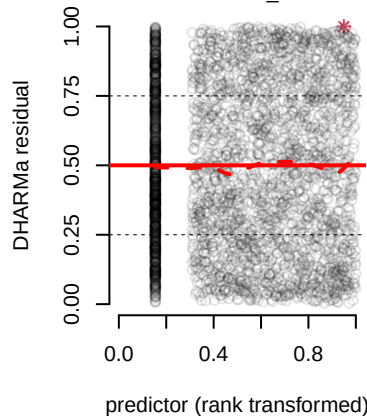
DHARMa residual Residual vs. pred

Predictor: sst_sd



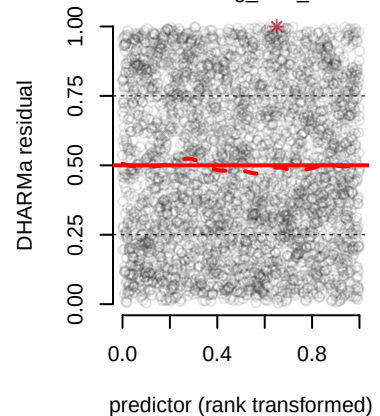
DHARMa residual Residual vs. pred

Predictor: sic_mean



DHARMa residual Residual vs. pred

Predictor: log_chla_mean



DHARMa residual Residual vs. pred

Predictor: depth

